Coding Assignment 9 | DATA 5600

Poisson Regression

Kolbee Morris

In [2]:
```
# load packages here
import pandas as pd
import numpy as np
import seaborn as sns
from matplotlib import pyplot as plt
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor as vif
from sklearn.linear_model import PoissonRegressor
from scipy.stats import chi2

# set seed
import random
random.seed(5600)

# function to perform the likelihood ratio test
def lr_test(model_full, model_reduced):
    dev_diff = model_reduced.deviance - model_full.deviance
    df_diff = model_reduced.df_resid - model_full.df_resid
    p_value = chi2.sf(dev_diff, df_diff)
    return {"Deviance Diff": dev_diff, "DF Diff": df_diff, "p-value": p_value}
```

Data and Description

This dataset helps estimate the recreational value of the Snake River in Wyoming—an important natural resource used for rafting, fishing, and other outdoor activities. Understanding this value can help policymakers, public land managers, and environmental economists make informed decisions about river access, conservation investments, or permit systems.

The analysis uses a travel cost model, a common economic method for valuing recreational sites. It assumes that the time and money people spend traveling to a site reflect how much they value visiting it. More trips generally indicate higher value.

Data were collected from:

1. An on-site intercept survey, where visitors were approached and surveyed as they left the river at key access points. This method captures people who are actively using the site, providing detailed information on recent trips and travel patterns.
2. A household mail survey sent to randomly selected Wyoming residents, which captures river use among the broader population.

The data set has information for 731 days and contains the following variables:

Variable	Description
anvisits	Number of annual visits to park
gender	Gender, categorical with 2 levels: (1) = male, (0) female
income	Income in US dollars per year, categorical with 4 levels: (1) <=$25,000$, (2) >25,000 - $55,000$, (3) >55,000 - $95,000$, (4) >95,000
travel	Travel time in hours, categorical with 3 levels: (1) <0.25 hours, (2) >=0.25 - <4 hours, (3) >=4 hours

The data can be found in the nationalPark data set on Canvas. Download nationalPark.csv, and put it in the same folder as this Jupyter notebook.

0. Replace the text "< PUT YOUR NAME HERE >" (above next to "author:") with your full name.

1. Read in the data set, and call the data frame "park". Print the fisrt few rows of the data.

In [3]:
```
park = pd.read_csv("nationalPark.csv")
park.head()
```

Out[3]:

	anvisits	gender	income	travel
0	2	1	4	3
1	1	1	4	3
2	1	0	4	2
3	1	0	4	3
4	1	1	3	3

2. Make a new data frame called "park_dummy" where the categorical variables are dummy coded. Note, that gender is already ready to go, so skip that variable. Print a summary of the data and make sure the data makes sense. (Hint: set `drop_first=True` in the `pd.get_dummies` function to remove one level of the categorical variables to avoid multicollinearity issues.)

In [4]:

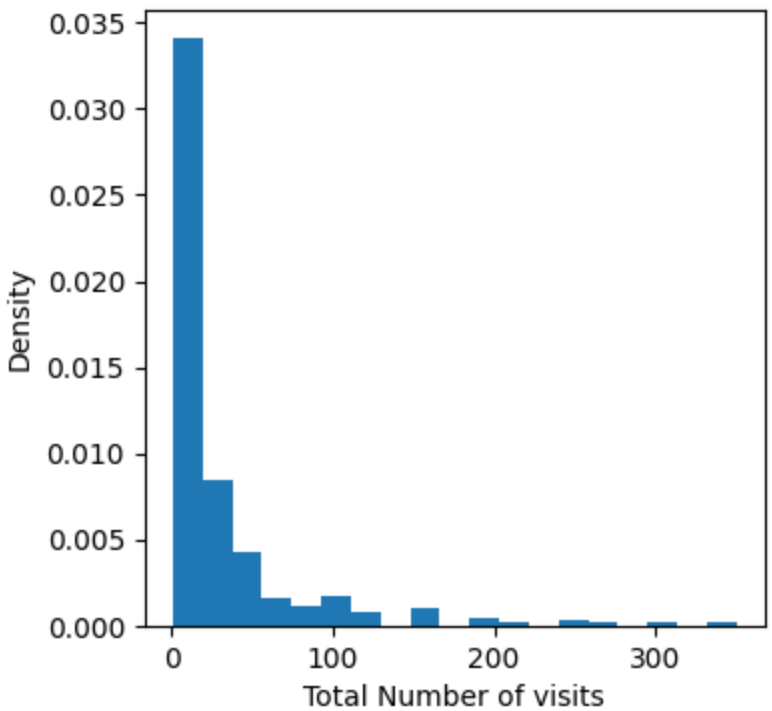
```
park_dummy = pd.get_dummies(park,
                            columns = ['income', 'travel'],
                            drop_first = True,
                               dtype = "int")
park_dummy.describe()
```

Out[4]:

	anvisits	gender	income_2	income_3	income_4	travel_2	travel_3
count	342.000000	342.000000	342.000000	342.000000	342.000000	342.000000	342.000000
mean	27.730994	0.377193	0.245614	0.228070	0.380117	0.377193	0.359649
std	48.955520	0.485394	0.431081	0.420203	0.486127	0.485394	0.480601
min	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	6.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	30.000000	1.000000	0.000000	0.000000	1.000000	1.000000	1.000000
max	350.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

3. Explore the data: create a histogram for the response. *Briefly describe the shape of the distribution - you should mention (1) symmetry or skewness, (2) the number of modes, and (3) potential outliers.*

In [5]:
```
fig = plt.figure(figsize = (4, 4))
plt.hist(park['anvisits'], density = True, bins = 19)
plt.xlabel("Total Number of visits")
plt.ylabel("Density")
plt.show()
```



This distribution is very right-skewed. almost all of the data is on that right side. having a long tail going out to the left. The plot is one mode. there is only one prominant peak and that is within the 0-20 visit range. I would say that most outliars are beyond the 200 number of visits anything then on is considered an outlier.

4. Briefly explain ONE reason why traditional multiple linear regression methods are not suitable for *this* data set (*your reasons should refer to this data set (i.e. be specific, not general)*).

Anvisits is a count varible that is right skewed, traditional multiple linear regression would assume that the response varible is contiunious and normally distributed. Since we are dealing with data that is non negative traditional MLR could produce data such as negative annual visits. Poisson is best because it is specifically designed for count responses.

5. Write out the general/theoretical Poisson regression model for this data set using all predictors. Be sure to use indicator variables as appropriate. You may need to split the equation on multiple lines to have it render properly as an HTML file.

$$\log(\lambda_i) = \beta_0 +$$

$$\beta_1 \text{gender}_i +$$

$$\beta_2 \text{I}(\text{income}_i = 2) +$$

$$\beta_3 \text{I}(\text{income}_i = 3) +$$

$$\beta_4 \text{I}(\text{income}_i = 4) +$$

$$\beta_5 \text{I}(\text{travel}_i = 2) +$$

$$\beta_6 \text{I}(\text{travel}_i = 3)$$

where λ_i is the average number of annual park visits for the ith person.

6. Fit an inital Poisson regression model using all predictor variables.

In [6]:
```
y = park_dummy[['anvisits']]
# Get the X design matrix ready
X_pois = park_dummy.filter(regex = '^gender|^income|^travel')
# use regular expressions to quickly include all dummy variables for to a variable
X_pois.head()
```

Out[6]:

	gender	income_2	income_3	income_4	travel_2	travel_3
0	1	0	0	1	0	1
1	1	0	0	1	0	1
2	0	0	0	1	1	0
3	0	0	0	1	0	1
4	1	0	1	0	0	1

In [7]:

```
X_pois = sm.add_constant(X_pois)
mod = sm.GLM(y,
             X_pois,
             family = sm.families.Poisson())
res = mod.fit()
res.summary()
```

Out[7]:

Generalized Linear Model Regression Results

Dep. Variable:	anvisits	**No. Observations:**	342
Model:	GLM	**Df Residuals:**	335
Model Family:	Poisson	**Df Model:**	6
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-6323.5
Date:	Wed, 15 Apr 2026	**Deviance:**	11325.
Time:	17:07:01	**Pearson chi2:**	1.84e+04
No. Iterations:	6	**Pseudo R-squ. (CS):**	1.000
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	4.0967	0.028	146.902	0.000	4.042	4.151
gender	0.3533	0.021	16.659	0.000	0.312	0.395
income_2	-0.0142	0.027	-0.534	0.594	-0.067	0.038
income_3	-0.5560	0.034	-16.197	0.000	-0.623	-0.489
income_4	-0.5640	0.031	-18.237	0.000	-0.625	-0.503
travel_2	-0.6188	0.022	-28.468	0.000	-0.661	-0.576
travel_3	-2.4394	0.049	-49.563	0.000	-2.536	-2.343

The next several questions involve using diagnostics to check the Poisson regression model assumptions. For each assumption, (1) code the diagnostic(s) that I indicate (next to the assumption in parentheses) to determine if the assumption is violated, and (2) explain whether or not you think the assumption is violated and why you think that. **Note: We do not have any continuous predictors, so we do not need to assess the linearity or "no multicollinearity" assumptions.**

7. The residuals are independent (no diagnostic tools - just think about how the data was collected and briefly write your thoughts)

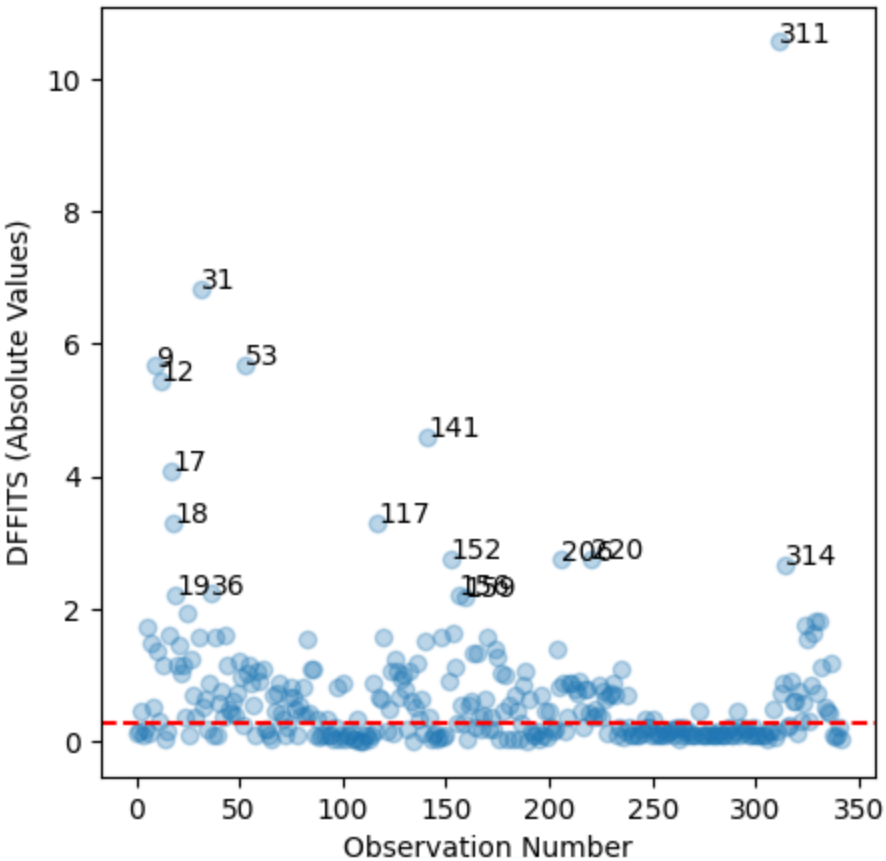
We dont explicitly know weather or not the results are independant. They randomly surveyed people who walked around the park and also sent out the survey randomly to wyoming residents. But there could be a likelyhood of them accidentally surveying the same person twice. I would assume they are independant but not certain.

8. The model describes all observations (i.e., there are no influential points) (use DFFITS)

In [8]:
```
# calculate the DFFITS
park['dffits'] = res.get_influence().summary_frame()['dffits_internal']

# plot DFFITS against the observation number
fig = plt.figure(figsize = (5, 5))
plt.ylabel("DFFITS (Absolute Values)")
plt.xlabel("Observation Number")
plt.scatter(park.index,
            np.abs(park['dffits']),
            alpha = 0.3)
plt.axhline(y = 2 * np.sqrt(len(res.params) / len(park)),
            color = 'r',
            linestyle = 'dashed')

# OPTIONAL: code to label potential outliers
# Note: I changed the cutoff since too many were labelled as outliers otherwise
outliers = park[np.abs(park['dffits']) > 2] # 2 * np.sqrt(len(res.params) / len(crash))]
for i in list(outliers.index):
        plt.annotate(str(i),
                     (i, np.abs(park['dffits'][i])))
plt.show()
```



There seems to be a lot of dots above the red line. Most of the dots seem to be close together as well. There is that one point 311 that is far a way from the other points. This will need to be check before doing anything else in the model. This assumption is not met until we check that point.

9. Additional predictor variables are not required (no diagnostic tools - just think about the variables you have and if there are other variables you think would help predict the response)

Demographics would be a great varible to add. This could help us see what ages and employment status most people have. this is huge because we can see what ages are visiting the parks the most.

10. Mean = Variance (no overdispersion/underdispersion) (use the three methods discussed in class)

In [9]:
```
# Compare the mean and variance
print(np.mean(park['anvisits']))
print(np.var(park['anvisits']))
```

```
27.730994152046783
2389.6352381929487
```

In [10]:
```
# Conduct the formal test
1 - chi2(df = res.df_resid).cdf(res.deviance)
```

Out[10]:
```
np.float64(0.0)
```

In [11]:
```
# Calculate dispersion parameter estimate
res.pearson_chi2 / res.df_resid
```

Out[11]:
```
np.float64(55.02964408328572)
```

the variance is roughly 86 times larger than the mean, which is a strong indication of overdispersion.

A p-value of essentially 0 confirms that the model is a significantly poor fit, consistent with overdispersion.

the dispersion parameter should be equal to 1. Here it is 55.03. meaning that the variance is 55 times larger than what the poisson model assumes.

Regardless of your assessment of the assumptions, proceed as if all assumptions were met for the sake of this assignment. Note that in "real life," you'd need to fit a different model to this data (you could try the negative binomial).

11. Use the likelihood ratio test to determine if the income predictor can be removed from the model. Note that researchers are interested in statistical inference and an explainable model. What do you conclude - can the income variable be dropped?

In [12]:
```
X_no_income = X_pois.drop(['income_2', 'income_3', 'income_4'], axis = 1)
X_no_income.head()
```

Out[12]:

	const	gender	travel_2	travel_3
0	1.0	1	0	1
1	1.0	1	0	1
2	1.0	0	1	0
3	1.0	0	0	1
4	1.0	1	0	1

In [13]:

```
mod_no_income = sm.GLM(y,
                       X_no_income,
                       family = sm.families.Poisson())
res_no_income = mod_no_income.fit()
res_no_income.summary()
```

Out[13]:

Generalized Linear Model Regression Results

Dep. Variable:	anvisits	**No. Observations:**	342
Model:	GLM	**Df Residuals:**	338
Model Family:	Poisson	**Df Model:**	3
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-6656.8
Date:	Wed, 15 Apr 2026	**Deviance:**	11992.
Time:	17:07:01	**Pearson chi2:**	2.22e+04
No. Iterations:	6	**Pseudo R-squ. (CS):**	1.000
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	3.8186	0.019	198.688	0.000	3.781	3.856
gender	0.4062	0.021	19.298	0.000	0.365	0.447
travel_2	-0.5841	0.022	-26.874	0.000	-0.627	-0.541
travel_3	-2.6284	0.049	-54.135	0.000	-2.724	-2.533

In [14]:

```
# Likelihood ratio test (function defined at the top of the notebook)
print("Full vs No income:", lr_test(res, res_no_income))
```

```
Full vs No income: {'Deviance Diff': np.float64(666.5470239808255), 'DF Diff': np.int64(3), 'p-value': np.f
loat64(3.764075579237188e-144)}
```

No income cannot be removed from the model. This is because of the deviance difference of 666 test statistic. This increase is huge. The penilty for dropping income is too huge to do so. keeping income gives us a better "explainable model".

12. Print a summary of your original model with all covariates for easy reference.

In [15]:

```
res.summary()
```

Out[15]:

Generalized Linear Model Regression Results

Dep. Variable:	anvisits	**No. Observations:**	342
Model:	GLM	**Df Residuals:**	335
Model Family:	Poisson	**Df Model:**	6
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-6323.5
Date:	Wed, 15 Apr 2026	**Deviance:**	11325.
Time:	17:07:01	**Pearson chi2:**	1.84e+04
No. Iterations:	6	**Pseudo R-squ. (CS):**	1.000
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	4.0967	0.028	146.902	0.000	4.042	4.151
gender	0.3533	0.021	16.659	0.000	0.312	0.395
income_2	-0.0142	0.027	-0.534	0.594	-0.067	0.038
income_3	-0.5560	0.034	-16.197	0.000	-0.623	-0.489
income_4	-0.5640	0.031	-18.237	0.000	-0.625	-0.503
travel_2	-0.6188	0.022	-28.468	0.000	-0.661	-0.576
travel_3	-2.4394	0.049	-49.563	0.000	-2.536	-2.343

13. Create (and output) 95% confidence intervals for the gender coefficient for $\exp\{\hat{\beta}_{\text{gender}}\}$ and $100 \times (\exp\{\hat{\beta}_{\text{gender}}\} - 1)$ using the `conf_int` function.

In [16]:

```
# Get the 95% CI for all coefficients
conf_int = res.conf_int(alpha = 0.05)
lrr_ci_gender = conf_int.loc['gender']
np.exp(lrr_ci_gender)
```

Out[16]:
```
0    1.365842
1    1.484252
Name: gender, dtype: float64
```

In [17]:
```
(np.exp(lrr_ci_gender) - 1) * 100
```

Out[17]:
```
0    36.584192
1    48.425226
Name: gender, dtype: float64
```

14. Interpret the 95% confidence intervals for gender for $\exp\{\hat{\beta}_{\text{gender}}\}$ and $100 \times (\exp\{\hat{\beta}_{\text{gender}}\} - 1)$ (two interpretations total).

We are 95% confident that holding all other predictors constantthe expected number of visits for males is between 1.366 and 1.484 times the expected number of visits for females.

We are 95% confident that holding all other predictors constant males have an expected number of visits that is between 36.58% and 48.43% higher than females.

15. Calculate (and output) a 95% confidence interval (and point estimate) for the predicted average number of annual visits to the park for people where gender = 0 (female), income = 2 (between $25,000 and$55,000), and travel = 3 (4 hours or more). *Interpret the interval.*

In [20]:
```
pred = res.get_prediction([1, 0, 1, 0, 0, 0, 1])  # const, gender=0, income_2=1, income_3=0, income_4=0, t
pred.summary_frame(alpha=0.05)
```

Out[20]:

	mean	mean_se	mean_ci_lower	mean_ci_upper
0	5.170925	0.262783	4.680698	5.712495

We are 95% confident that for females who average between 25,000 and 55,000 dollars that travel 4 or more hours the true mean number of annual visits for all people in this group falls between 4.68 and 5.71 visits per year.

16. Compute (and output) the likelihood ratio test statistic for the model, and compute (and output) the associated p-value. Based on the results, what do you conclude?

In [21]:
```
res.summary()
```

Out[21]:

Generalized Linear Model Regression Results

Dep. Variable:	anvisits	**No. Observations:**	342
Model:	GLM	**Df Residuals:**	335
Model Family:	Poisson	**Df Model:**	6
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-6323.5
Date:	Wed, 15 Apr 2026	**Deviance:**	11325.
Time:	17:16:17	**Pearson chi2:**	1.84e+04
No. Iterations:	6	**Pseudo R-squ. (CS):**	1.000
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	4.0967	0.028	146.902	0.000	4.042	4.151
gender	0.3533	0.021	16.659	0.000	0.312	0.395
income_2	-0.0142	0.027	-0.534	0.594	-0.067	0.038
income_3	-0.5560	0.034	-16.197	0.000	-0.623	-0.489
income_4	-0.5640	0.031	-18.237	0.000	-0.625	-0.503
travel_2	-0.6188	0.022	-28.468	0.000	-0.661	-0.576
travel_3	-2.4394	0.049	-49.563	0.000	-2.536	-2.343

In [22]:
```
# Test statistic
like_ratio = res.null_deviance - res.deviance
print('test-statistic =', like_ratio)
# p-value from a chi-square distribution with the
# number of predictors as the degrees of freedom
print('p-value =', chi2.sf(like_ratio, res.df_model))
```

```
test-statistic = 7304.464315678037
p-value = 0.0
```

With a test-statistic of 7304.46 and a p-value of 0 we would reject the null hypothesis. at least one predictor varible (gender, income, travel) has a significant relationship with the number of annual visits a person has at the park.

17. Compute (and output) the pseudo R^2 value for the model.

In [ ]:
```
1 - (res.deviance / res.null_deviance)
```

18. Briefly summarize what you learned from this analysis *to a non-statistician*. Write a few sentences about (1) the purpose of this data set and analysis and (2) what you learned about this data set from your analysis. Write your response as if you were addressing a business manager (avoid using statistics jargon) and just provide the main take-aways.

Here we wanted to see what the average visits were for people from different backgrounds such as gender income and how far they have to travel. The goal was to identify which characteristics most influenced visitor loyalty and frequency. We found that gender and income were the primary drivers in predicting why someone came to the parks. In short it is not random there are clear predictors in who come to the parks from gender, income, and travel standpoints.

In [ ]: